

TDMA-CBBA: Communication-Constrained Task Allocation for Drone Swarms with Provable Convergence Guarantees

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Abstract—Decentralized task allocation is a cornerstone of autonomous drone swarm operations, and the Consensus-Based Bundle Algorithm (CBBA) remains its most widely deployed solution, offering convergence guarantees and a 50% optimality bound under idealized, instantaneous, all-to-all communication. Real swarms, however, share a contention-limited radio channel: uncoordinated transmissions collide, packets are lost, and information propagates with bounded but nonzero delay. We present TDMA-CBBA, a communication-constrained extension of CBBA in which agents transmit exclusively within Time Division Multiple Access (TDMA) slots, subject to stochastic packet loss, and maintain time-stamped belief ledgers that expire after a validity horizon T_{val} . Our central theoretical result identifies the exact operating regime for guaranteed convergence: CBBA’s consensus dynamics are preserved under TDMA if and only if one full TDMA frame fits inside the belief-validity horizon ($N\tau \leq T_{\text{val}}$). Within this regime we prove that (i) consensus terminates within $T_{\text{TDMA}} \leq T_{\text{ideal}} + N\tau K$ for K auction rounds, and (ii) the 50% optimality guarantee of CBBA is preserved despite stale bids. Outside this regime, we show empirically that expired beliefs trigger cascading allocation erasures that provably prevent consensus—a previously unreported *synchronization-staleness phase transition*. We validate our analysis on an open-source benchmark of 1,620 simulation runs across 54 configurations spanning swarm sizes of 10–50 drones, packet loss rates up to 30%, and four baseline radio models, quantifying trade-offs between mean time to consensus (31.26 s, 95% CI [30.22, 32.30]), coverage efficiency (0.76), and per-link packet delivery ratio (0.94 at mixed loss). Code, data, and analysis scripts are publicly released.¹

I. INTRODUCTION

Autonomous drone swarms promise transformative capabilities in search and rescue, environmental monitoring, infrastructure inspection, and defense [4], [5]. In all of these missions, a central computational problem recurs: *which robot should do which task?* Centralized assignment is brittle—it introduces a single point of failure, a communication bottleneck at the leader, and a scalability ceiling that grows with fleet size [2], [3]. The field’s dominant answer is the **Consensus-Based Bundle Algorithm** (CBBA) of Choi, Brunet, and How [1], a decentralized auction that provably converges to a conflict-free, 50%-optimal assignment for scoring functions satisfying

the Diminishing Marginal Gains (DMG) property. CBBA and its descendants now underpin task allocation in hundreds of multi-robot systems [7], [8].

A. Motivation: The Communication Gap

CBBA’s guarantees, however, rest on an implicit physical assumption: that agents exchange their bid ledgers *perfectly, instantaneously, and simultaneously*. No real swarm satisfies this assumption. A fleet of N drones sharing one half-duplex radio channel cannot transmit concurrently; uncoordinated transmissions collide; packets are dropped by fading and interference; and information about a teammate’s bid arrives only after a nonzero, topology-dependent delay [15], [16]. In contested environments—the very settings where swarms are most valuable—adversarial jamming and range limits widen this gap further.

Closing it naively is dangerous. Serializing CBBA’s message exchanges onto a shared channel introduces *information staleness*: each agent acts on beliefs about teammates that are outdated by up to one full communication frame, and stale beliefs interact destructively with CBBA’s conflict-resolution logic. Worse, when practitioners add the standard engineering safeguard of *belief expiry*—discarding telemetry older than a validity horizon T_{val} , so that lost or departed agents are forgotten [16]—we find that staleness can become fatal: if a full TDMA cycle $N\tau$ exceeds T_{val} , every agent’s belief about every teammate expires before it can be refreshed; cascading allocation erasures then race against re-learning, and consensus becomes *structurally unreachable*. To our knowledge, this failure mode—a *synchronization-staleness phase transition* governed by the ratio $N\tau/T_{\text{val}}$ —has not been identified in the CBBA literature, despite belief-expiry heuristics being common practice in deployed systems.

B. Our Approach

We propose **TDMA-CBBA**, a communication-constrained formulation of CBBA in which: (i) agents transmit exclusively within assigned Time Division Multiple Access slots of duration τ , eliminating intra-frame collisions by construction; (ii) transmissions are subject to independent stochastic packet loss

¹<https://github.com/721189/swarm-robotics>

on both transmit and receive legs; (iii) each agent maintains a time-stamped ledger of peers' bids and allocations, filtered through a belief-validity horizon T_{val} ; and (iv) a dynamic slot-allocation layer (D-TDMA) lets agents request doubled transmission time near threats or objectives, prioritizing safety-critical updates. Rather than treating the radio as an afterthought, we place the interplay between scheduling delay ($N\tau$) and belief staleness (T_{val}) at the center of both the analysis and the experimental design.

C. Contributions

This paper makes four contributions:

- 1) **Convergence theory under scheduled, lossy communication (Thm. 1).** We prove that TDMA-CBBA converges to a conflict-free assignment within $T_{TDMA} \leq T_{ideal} + N\tau K$ for K auction rounds, provided one full TDMA frame fits within the belief-validity horizon, $N\tau \leq T_{\text{val}}$ (Lemma 1). To our knowledge this is the first convergence result for CBBA under slotted medium access with explicit belief expiry.
- 2) **Optimality preservation despite stale bids (Thm. 3).** We prove that CBBA's 50% optimality guarantee survives stale-bid conflicts induced by bounded communication delay: monotone ledger merging ensures transient misallocations are repaired by subsequent consensus rounds, so the final assignment retains $\mathcal{U}_{TDMA} \geq 0.5\mathcal{U}$ for DMG score functions.
- 3) **A synchronization–staleness phase transition (Sec. IV-C, Thm. 2).** We prove that when $N\tau > T_{\text{val}}$, belief expiry renders global consensus unreachable, and we validate this impossibility result empirically: trials beyond the transition exhibit consensus rates collapsing toward zero while sub-transition trials converge reliably. This yields an actionable design rule—scale the validity horizon with the TDMA frame.
- 4) **An open-source benchmark with rigorous statistics.** We release a fully deterministic headless benchmark (1,620 runs across 54 configurations; swarm sizes 10–50, packet loss up to 30%, three slot durations, four baseline radio models), formal metric definitions, non-parametric statistical analysis, and five publication-quality figures. All code, raw data, and analysis pipelines are public.²

D. Paper Outline

Section II formulates the swarm, task, and communication models. Section III details the TDMA-CBBA methodology, including the D-TDMA extension. Section IV presents the theoretical analysis: convergence under ideal communication, convergence and the phase transition under TDMA constraints, worst-case optimality, and complexity. Section V specifies the experimental protocol. Section VI reports results. Section VII discusses limitations and future work, and Section VIII concludes.

TABLE I
SUMMARY OF NOTATION.

Symbol	Meaning
N, M	Number of drones, number of objectives
$\mathbf{p}_i(t), \mathbf{v}_i(t)$	Position, velocity of drone i at time t
$a_i(t) \in \mathcal{O} \cup \{\emptyset\}$	Task assignment of drone i
$b_i(t) \in [0, B_{\text{max}}]$	Battery level of drone i
$s_i \in \{0, \dots, N-1\}$	TDMA slot index of drone i
τ	TDMA slot duration [s]
$p_{\text{tx}}, p_{\text{rx}}$	Transmit-, receive-leg packet loss probability
c_{ij}	Bid (score) of drone i for objective j
$\mathcal{L}_i(t) = (\mathcal{W}_i, \mathcal{B}_i)$	Ledger i : winner map $\mathcal{W}_i$, bid map $\mathcal{B}_i$
T_{val}	Belief-validity horizon [s]
K	Number of auction/consensus rounds
L	Workspace side length (100m)

II. PROBLEM FORMULATION

We consider a team of N homogeneous drones required to allocate themselves to a set of geographically distributed objectives in a bounded planar workspace, under a contention-limited radio channel. Table I summarizes the notation.

A. Swarm Model

Let $\mathcal{D} = \{d_1, d_2, \dots, d_N\}$ be a set of N autonomous drones operating in a bounded two-dimensional environment $\mathcal{W} \subset \mathbb{R}^2$, a square of side $L = 100$ m. Each drone d_i is characterized at time t by:

- position $\mathbf{p}_i(t) \in \mathcal{W}$ and velocity $\mathbf{v}_i(t) \in \mathbb{R}^2$, integrated at timestep $\Delta t = 0.1$ s with first-order steering dynamics;
- battery level $b_i(t) \in [0, B_{\text{max}}]$; a drone whose battery reaches zero is removed from $\mathcal{D}$ (attrition);
- task assignment $a_i(t) \in \mathcal{O} \cup \{\emptyset\}$;
- a local *ledger* $\mathcal{L}_i(t) = (\mathcal{W}_i(t), \mathcal{B}_i(t))$, where $\mathcal{W}_i : \mathcal{O} \rightarrow \mathcal{D} \cup \{\perp\}$ maps each objective to its believed winner and $\mathcal{B}_i : \mathcal{O} \rightarrow \mathbb{R}_{\geq 0}$ stores the corresponding highest known bid;
- a *mailbox* $\mathcal{M}_i(t) = \{(j, \mathbf{m}, t^{\text{rx}})\}$ of time-stamped messages received from peers.

B. Objective and Threat Model

Let $\mathcal{O} = \{o_1, o_2, \dots, o_M\}$ be a set of M static objectives. Each objective o_j has position $\mathbf{q}_j \in \mathcal{W}$ and value $v_j \in \mathbb{R}^+$. The environment additionally contains k surface-to-air-missile (SAM) threat zones $\mathcal{T} = \{\theta_1, \dots, \theta_k\}$; each threat θ_l is a disc $(\mathbf{x}_l, r_l, s_l)$ centered at $\mathbf{x}_l$ with radius r_l and strength s_l that exerts on drone i a repulsive potential

$$U_{\text{threat}}(\mathbf{p}_i) = s_l \left(\frac{1}{d(\mathbf{p}_i, \mathbf{x}_l)} - \frac{1}{r_l} \right)^2, \quad d(\mathbf{p}_i, \mathbf{x}_l) < r_l, \quad (1)$$

where $d(\cdot, \cdot)$ denotes Euclidean distance. Threats shape motion but do not directly destroy drones in this study; agent loss occurs only through battery attrition.

C. Communication Model

Communication is governed by a slotted medium-access discipline with stochastic loss:

²<https://github.com/721189/swarm-robotics>

(C1) *Slotted access (TDMA)*. Time is divided into slots of duration τ ; each drone d_i owns exactly one slot $s_i \in \{0, 1, \dots, N-1\}$ per frame of length $N\tau$, and may broadcast only during s_i . A frame therefore contains exactly one transmission opportunity per drone.

(C2) *Two-stage packet loss*. A broadcast by d_i is lost entirely with probability p_{tx} ; independently, each intended receiver d_j fails to decode the packet with probability p_{rx} . Delivery on link (i, j) thus succeeds with probability $(1 - p_{tx})(1 - p_{rx})$.

(C3) *Belief validity horizon*. Each received message is stored with its timestamp and is considered valid only while $t - t^{msg} \leq T_{val}$. Expired beliefs are purged from $\mathcal{M}_i$, which in turn removes the corresponding peer from drone i 's *perceived swarm*—the set of teammates it believes to be alive and active.

(C4) *Baselines*. Setting $\tau \rightarrow 0$ (all agents transmit every frame) yields an unslotted *open channel*; further setting $p_{tx} = p_{rx} = 0$ yields the *ideal* channel assumed by canonical CBBA [1]. These degenerate cases define the four radio baselines evaluated in Section V.

D. Task Allocation Problem Statement

The objective is a decentralized, conflict-free assignment $a : \mathcal{D} \rightarrow \mathcal{O} \cup \{\emptyset\}$ that maximizes total utility

$$\mathcal{U}(a) = \sum_{i=1}^N c_{ii}(a(i)), \quad (2)$$

where $c_{ij} = \max(0.1, C_0 - d(\mathbf{p}_i, \mathbf{q}_j))$ is drone i 's bid for objective o_j ($C_0 = 100$), subject to (a) each objective receiving at most one winner, (b) convergence to ledger agreement within the mission horizon, and (c) operation over the lossy slotted channel of Section II-C. Conflict-freeness is defined by agreement: the allocation is settled when all ledgers coincide, $\mathcal{L}_1(t) = \dots = \mathcal{L}_N(t)$.

III. METHODOLOGY: TDMA-CBBA

TDMA-CBBA interleaves CBBA's two-phase consensus-auction loop with a slotted radio discipline and belief management. Algorithm 1 summarizes one simulation frame; the components relative to canonical CBBA—slotted medium access, two-stage packet loss, and belief expiry—are detailed in Sections III-C–III-E, followed by the D-TDMA extension in Section III-F.

A. Phase I: Consensus Auction over Ledgers

Each drone executes the CBBA phases locally, restricted to its perceived swarm $\mathcal{P}_i$:

(a) *Ghost-drone pruning*. Before bidding, drone i removes from its ledger every objective whose believed winner $w \notin \mathcal{P}_i \cup \{i\}$: it resets $\mathcal{B}_i(o_j) \leftarrow 0$ and $\mathcal{W}_i(o_j) \leftarrow \perp$, releasing the allocation. This mechanism provides fault tolerance against departed agents, but as Section IV-C shows, it is also the mechanism through which belief expiry can destabilize consensus.

(b) *Local greedy bid*. Drone i evaluates each objective o_j whose score satisfies $c_{ij} > \mathcal{B}_i(o_j)$ and claims the objective with maximal net gain $c_{ij} - \mathcal{B}_i(o_j)$, updating $a_i, \mathcal{B}_i(o_j) \leftarrow c_{ij}$,

Algorithm 1 TDMA-CBBA frame execution (one tick, Δt)

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1:  $t \leftarrow t + \Delta t$ 
    $\triangleright$  // Phase 1: Radio arbitration
2: if slot discipline active then
3:    $s \leftarrow \text{TDMA Scheduler.currentSpeaker}(t)$ 
4:   if  $d_s$  alive and  $\text{rand}() > p_{tx}$  then
5:     enqueue  $(s, d_s.\text{prepareBroadcast}(t))$ 
6:   end if
7: else
8:   for all alive  $d_i$  with  $\text{rand}() > p_{tx}$  do
9:     enqueue  $(i, d_i.\text{prepareBroadcast}(t))$ 
10:  end for
11: end if
    $\triangleright$  // Phase 2: Reception with per-link loss
12: for all enqueued  $(i, \mathbf{m}_i)$  and alive receivers  $d_j, j \neq i$  do
13:   if  $\text{rand}() > p_{rx}$  then
14:      $d_j.\text{receive}(i, \mathbf{m}_i)$ ; update PDR counters
15:   end if
16: end for
    $\triangleright$  // Phase 3: Belief filtering and agent step
17: for all alive  $d_i$  do
18:    $\mathcal{P}_i \leftarrow \{j : t - t_{ij}^{msg} \leq T_{val}\}$   $\triangleright$  perceived swarm
19:    $d_i.\text{auctionAndMove}(\mathcal{P}_i)$ 
20: end for
21: advance scheduler clock by  $\Delta t$ 

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and $\mathcal{W}_i(o_j) \leftarrow i$; any previously held task is released when switching.

(c) *Ledger merging on reception*. Upon receiving a broadcast from peer k , drone i performs an element-wise monotone merge: if the incoming bid exceeds its own for some o_j , it adopts both that bid and the sender's winner entry. If the adopted winner displaces i 's own assignment ($a_i = o_j$ but $\mathcal{W}_i(o_j) \neq i$), the outbid drone immediately releases $a_i \leftarrow \emptyset$.

Because merges are monotone in bid value and winner entries travel with bids, this update rule is exactly the max-consensus of CBBA [1]; Lemma 2 formalizes the property driving our convergence proofs.

B. Phase II: Motion Control

Between auctions, drones execute a potential-field controller: attraction toward the assigned objective, repulsion from SAM threat zones via Eq. (1), and Reynolds-style separation from neighbors within 4 m—where neighbors are drawn from the perceived swarm $\mathcal{P}_i$ under TDMA (strictly local information), or from ground truth under the open-channel baselines. This asymmetry is deliberate: it makes motion itself sensitive to communication quality, so that SCI responds to both physical flocking and informational coupling.

C. Slotted Medium Access

Under TDMA, exactly one agent broadcasts per scheduling opportunity, eliminating intra-frame collisions by construction. The scheduler advances round-robin through slots, so each agent transmits once per frame of duration $N\tau$; an agent's

maximum *inter-transmission gap* equals $N\tau$. This gap is precisely the quantity that couples to the belief horizon T_{val} in Section IV: whenever the gap exceeds T_{val} , the freshest belief about that peer expires before it can be renewed. In the open-channel baselines every alive agent broadcasts every frame, gaps collapse to one tick, and staleness becomes negligible.

D. Two-Stage Packet Loss Model

We model channel impairment at two physically distinct points: (i) *transmit-leg loss* p_{tx} , representing collisions with external traffic, emitter-side fades, or aborted transmissions; and (ii) *receive-leg loss* p_{rx} , representing per-receiver decode failures due to distance-dependent SNR. In our benchmark $p_{\text{rx}} = p_{\text{tx}}/2$ (the receiver leg benefits from coding gain), giving link success probability $(1-p_{\text{tx}})(1-p_{\text{tx}}/2)$. Both stages are independent across links and frames, and every delivery attempt is counted, enabling exact offline computation of PDR.

E. Belief Expiry (Validity Horizon)

Each mailbox entry carries its receive timestamp; entries older than T_{val} are purged before the auction phase. Belief expiry serves two purposes: it bounds mailbox memory, and it implements graceful degradation under agent attrition (departed agents stop transmitting and are forgotten within T_{val}). Its interaction with TDMA scheduling is the central object of Section IV: expiry converts bounded delay into bounded *belief age*, and consensus survives only while worst-case belief age stays below the horizon.

F. Dynamic TDMA (D-TDMA) Extension

Fixed slot lengths waste capacity on quiet agents and starve safety-critical ones. We therefore extend the scheduler with demand-driven slot doubling:

- 1) Each drone continuously evaluates the proximity predicate

$$\phi_i = \left(\min_l d(\mathbf{p}_i, \mathbf{x}_l) < r_l + 5 \text{ m} \right) \vee \left(\min_j d(\mathbf{p}_i, \mathbf{q}_j) < 20 \text{ m} \right), \quad (3)$$

i.e., the drone is near a threat or approaching an objective.

- 2) If ϕ_i holds, the drone raises a double-slot request flag addressed to the scheduler.
- 3) When the drone's next scheduled slot begins, the scheduler grants a doubled slot of length 2τ , then clears the flag.
- 4) Grants are non-preemptive and preserve round-robin order: at most one doubling occurs per visit, slot order is never permuted, so the frame remains collision-free and the analysis of Section IV carries over with effective cycle time bounded by $2N\tau$.

D-TDMA thus adds an adaptive, safety-aware layer on top of deterministic medium access: contested agents obtain up to twice the telemetry bandwidth exactly when their state estimates matter most (threat evasion, terminal approach), while Theorems 1 and 3 remain valid under the relaxed condition $2N\tau \leq T_{\text{val}}$.

IV. THEORETICAL ANALYSIS

This section develops the paper's formal guarantees. Section IV-A recalls the classical CBBA result; Section IV-B extends convergence to TDMA with belief expiry, culminating in Theorem 1 and the impossibility result of Section IV-C; Section IV-D proves optimality preservation under stale bids; Section IV-E analyzes complexity.

A. Convergence Under Ideal Communication

Choi *et al.* [1] proved that CBBA converges to a conflict-free assignment under two assumptions: (A1) the score function satisfies Diminishing Marginal Gains (DMG), and (A2) communication is complete and instantaneous—every pair of agents synchronizes ledgers every round. Under (A1)–(A2), consensus over K bundle-building rounds terminates with all agents agreeing on a single winner per objective.

The distance-based score used here, $c_{ij} = \max(0.1, C_0 - d(\mathbf{p}_i, \mathbf{q}_j))$, is monotone non-increasing in distance and satisfies DMG trivially for single-task claims (each agent holds at most one task), so (A1) holds throughout.

B. Convergence Under TDMA Constraints

We now relax (A2) to the slotted lossy channel of Section II-C. Two definitions make “eventual consistency” precise.

Definition 1 (Eventual ledger consistency): The ledgers $\{\mathcal{L}_i(t)\}_{i=1}^N$ are *eventually consistent* if there exists finite T_{cons} such that for all $t \geq T_{\text{cons}}$ and all i, j : $\mathcal{L}_i(t) = \mathcal{L}_j(t)$ —every drone holds identical winner and bid maps for all objectives.

Definition 2 (Bounded delivery delay): Communication is *bounded-delay* with bound Δ_{max} if every message generated at time t by any agent is delivered to every other agent no later than $t + \Delta_{\text{max}}$ or is lost.

Lemma 1 (TDMA frame as bounded delay): Under slotted access (C1) with losses $p_{\text{tx}}, p_{\text{rx}} < 1$, TDMA-CBBA is bounded-delay with

$$\Delta_{\text{max}} = 2N\tau, \quad (4)$$

provided each agent's freshest information survives at least one full frame:

$$N\tau \leq T_{\text{val}}. \quad (5)$$

Proof: Round-robin order guarantees that any agent's consecutive transmissions are spaced exactly $N\tau$ apart, so a receiver that misses a packet obtains a refresh within at most $N\tau$ more seconds; worst-case information age at any receiver therefore never exceeds one frame plus one slot rotation, i.e., $2N\tau$. Condition (5) ensures this age stays within the validity horizon, so beliefs are refreshed before they expire rather than purged. With $p_{\text{tx}}, p_{\text{rx}} < 1$, the probability that all successive attempts on a given link fail tends to zero geometrically over frames, so delivery is almost surely achieved within the stated bound. $\square$

Lemma 2 (Monotone ledger merging): For every objective o_j and agent i , the stored bid $\mathcal{B}_i^{(t)}(o_j)$ is non-decreasing in t except when reset by ghost pruning, and merging never

decreases either party's bid. Moreover $\mathcal{W}_i(o_j)$ always names the agent holding the bid $\mathcal{B}_i(o_j)$.

Proof: Both the reception merge (adopt only strictly greater bids) and the self-bid update (claim only when $c_{ij} > \mathcal{B}_i(o_j)$) write values strictly greater than the stored value; winner entries are copied together with bids from the same packet, so bid-winner pairs remain consistent. Ghost pruning resets a bid to 0 only when its believed winner has left the perceived swarm—which under condition (5) cannot be caused by scheduling alone, only by genuine attrition. $\square$

Theorem 1 (Convergence of TDMA-CBBA): Assume DMG scores (A1), slotted access (C1), losses $p_{tx}, p_{rx} < 1$, belief horizon T_{val} , and feasibility (5). Then TDMA-CBBA achieves eventual ledger consistency in bounded time

$$T_{TDMA} \leq T_{ideal} + N\tau K, \quad (6)$$

where T_{ideal} is canonical CBBA's convergence time on the same instance and K its required auction rounds. At termination the assignment is conflict-free.

Proof: Under ideal communication CBBA converges after K rounds at time T_{ideal} [1]. Under TDMA satisfying Lemma 1, every round- k broadcast reaches all peers almost surely within $2N\tau$, and by Lemma 2 each delivered packet makes monotone progress toward the ideal fixed point; no update can be undone because ghost pruning is inactive absent true attrition. Hence after at most K logical rounds—each stretched by at most one frame $N\tau$ of scheduling delay—the system attains the ideal fixed point, giving (6). Conflict-freeness follows from bid-winner consistency of Lemma 2: each objective has exactly one winner id recorded by all agents. $\square$

C. The Synchronization–Staleness Phase Transition

The feasibility condition (5) is not merely technical—crossing it flips the system's qualitative behavior.

Theorem 2 (Impossibility beyond the staleness threshold): If $N\tau > T_{val}$, then for every agent i and every peer $j \neq i$, the belief about j expires before j 's next scheduled transmission. Consequently each agent's perceived swarm shrinks to $\{i\}$ between consecutive receptions, ghost pruning erases all foreign winner entries every auction round, and no allocation can persist across a full frame: global consensus (Def. 1) is unreachable for any $N \geq 2$, regardless of runtime.

Proof: Agent d_j broadcasts once per frame, spaced exactly $N\tau$ apart. If $N\tau > T_{val}$, then just before d_j speaks again, every earlier packet from d_j in i 's mailbox has age exceeding T_{val} and is purged by (C3). Thus $\mathcal{P}_i$ contains no trace of j at auction time, and ghost pruning resets $\mathcal{B}_i(o_k) \leftarrow 0$, $\mathcal{W}_i(o_k) \leftarrow \perp$ for every objective believed won by j ; symmetrically, j erases i 's wins. Any allocation recorded in one ledger is therefore destroyed in the others within one frame, and re-established allocations are destroyed again in the next frame—the joint ledger state cycles without agreement, deterministically, for all $N \geq 2$. $\square$

Theorem 2 converts an engineering heuristic (belief expiry) into a hard scalability constraint: *the validity horizon must*

scale linearly with swarm size, $T_{val} \geq N\tau$. Section VI validates the predicted phase transition empirically.

D. Worst-Case Performance Guarantee

A remaining concern is whether stale bids—decisions made on information up to $2N\tau$ old—can degrade solution quality below CBBA's celebrated 50% bound.

Theorem 3 (Optimality preservation under stale bids): Let $\mathcal{U}^*$ be the optimal conflict-free utility and $\mathcal{U}_{TDMA}$ the utility at TDMA-CBBA termination. Under Theorem 1's conditions with DMG scores,

$$\mathcal{U}_{TDMA} \geq 0.5\mathcal{U}^*. \quad (7)$$

Proof: (i) *Staleness affects timing, not the score.* Bids depend only on positions, which evolve independently of the channel; a bid formed at $t - \Delta$ ($\Delta \leq 2N\tau$) is a valid DMG score for the configuration at which it was computed. (ii) *Conflicts are transient, not absorbing.* Suppose stale information leads two agents i, k to believe they win the same objective o_j , holding bids $\mathcal{B}_i(o_j)$ and $\mathcal{B}_k(o_j)$. Their next mutual ledger exchange—guaranteed within $2N\tau$ by Lemma 1—causes both to adopt the larger bid and its owner as sole winner, and the outbid agent releases its assignment by rule (c) of Sec. III-A. Every conflict is thus resolved within one frame of its creation, and none persists at termination. (iii) *The terminal state is a CBBA fixed point.* At consensus (guaranteed by Thm. 1), all agents agree on one winner per objective carrying the maximum bid seen anywhere in the network—precisely ideal CBBA's terminal condition restricted to the realized bid sequence. Because ideal CBBA achieves $\geq 0.5\mathcal{U}^*$ on DMG scores regardless of bid arrival order [1], and TDMA-CBBA realizes some permutation of a valid bid sequence, Eq. (7) follows. $\square$

Remark. Theorem 3 concerns the *final* allocation; intermediate assignments may transiently fall below the bound while conflicts await resolution. Our experiments measure convergence and coverage rather than realized utility, so Eq. (7) is validated structurally (conflict-freeness at consensus); adding a Hungarian-optimal comparator is identified as future work in Sec. VII.

E. Complexity Analysis

Table II compares per-agent costs, where $n = |\mathcal{D}|$, $m = |\mathcal{O}|$, K auction rounds, and one TDMA frame consumes $\lceil N\tau/\Delta t \rceil$ ticks.

Three observations follow. First, *computational* cost per agent per round is unchanged by TDMA: auctions and merges touch only the m -dimensional ledger, and belief filtering is linear in mailbox size. Second, *communication volume per agent drops from $O(n)$ point-to-point syncs to exactly one broadcast per round*, since a single packet carries the full ledger to all listeners simultaneously. Third, the price of slotted access is purely *temporal*: wall-clock per round inflates from one tick to one frame $N\tau$ —exactly the $N\tau K$ term of Theorem 1. Total network traffic per frame is $O(nm)$ bits in both cases; TDMA redistributes rather than amplifies traffic, trading latency for determinism and collision freedom.

TABLE II
PER-AGENT COMPUTATIONAL, MEMORY, AND COMMUNICATION
COMPLEXITY. TDMA'S OVERHEAD IS TEMPORAL (SERIALIZED SLOTS),
NOT EXTRA MESSAGES OR COMPUTE.

	Ideal CBBA	TDMA-CBBA
Auction compute / round	$O(m)$	$O(m)$
Ledger merge per message	$O(m)$	$O(m)$
Belief filtering per tick	—	$O(n)$
Memory (ledger + mailbox)	$O(nm)^\dagger$	$O(m+n)$
Messages sent / agent / round	$O(n)$	1
Wall-clock per round	$O(\Delta t)$	$O(N\tau)$

[†]Canonical CBBA stores full path bundles; ours stores flat ledgers.

V. EXPERIMENTAL SETUP

This section specifies the simulation platform, test matrix, metric definitions, baseline radio models, and statistical methodology. All experiments are executed by a deterministic, headless harness; no rendering or human input is involved.

A. Simulation Platform

The environment is a 100×100 m square workspace $\mathcal{W}$ with reflecting boundaries. Drones are homogeneous combat agents ($v_{\max} = 1.5$ m/s, steering limit 0.2, communication-sensing radius 25 m) integrated at $\Delta t = 0.1$ s for a mission horizon of $F_{\max} = 500$ frames (50 s). Objectives are placed uniformly at random in the central region $[15, 85]^2$; SAM threats (radius 15 m, strength 60) are placed uniformly in $[20, 80]^2$. Belief validity is fixed at $T_{\text{val}} = 2.0$ s throughout, so by Theorem 2 the predicted staleness threshold sits at swarm sizes where $N\tau > 2$ s—a regime our matrix deliberately spans.

B. Test Matrix

The released benchmark varies five factors: swarm size $n \in \{10, 30, 50\}$, packet loss $p_{\text{tx}} \in \{0.0, 0.05, 0.15, 0.30\}$ (with $p_{\text{rx}} = p_{\text{tx}}/2$), objectives $M \in \{3, 5, 10\}$, threats $k \in \{0, 2, 5\}$, and slot duration $\tau \in \{25, 50, 100\}$ ms. Each of the $3 \times 4 \times 3 \times 3 \times 3 = 324$ configurations is repeated over t independent trials with deterministic seeding (base seed 42, trial seed $42 + t'$), yielding fully reproducible runs. The dataset analyzed in Section VI uses $t = 5$ trials per configuration (1,620 runs); the identical harness supports arbitrary trial counts, including a 30-trial paper-grade grid (9,720 runs), via a single command-line flag.

Because $T_{\text{val}} = 2$ s and $\tau \leq 100$ ms, configurations with $n \geq 21$ at $\tau = 100$ ms ($N\tau > 2$ s) fall beyond the staleness threshold of Theorem 2; this lets us probe both sides of the phase transition within one matrix.

C. Metrics

Four metrics are recorded per trial. Let $\mathcal{D}_{\text{alive}}(t)$ denote alive agents at trial end and $L = 100$ m.

(1) *Swarm Cohesion Index (SCI)*—normalized mean pair-wise separation among alive agents:

$$\text{SCI} = \frac{2}{|\mathcal{D}_{\text{alive}}|(|\mathcal{D}_{\text{alive}}| - 1)} \sum_{i < j} \frac{\|\mathbf{p}_i - \mathbf{p}_j\|}{L}, \quad (8)$$

with $\text{SCI} \rightarrow 0$ indicating tight clustering and larger values dispersion; single-agent trials report $\text{SCI} = 1$.

(2) *Coverage Efficiency (CE)*—fraction of objectives claimed by an assigned, alive agent:

$$\text{CE} = \min\left(1, \frac{|\{d_i \in \mathcal{D}_{\text{alive}} : a_i \neq \emptyset\}|}{M}\right). \quad (9)$$

(3) *Packet Delivery Ratio (PDR)*—per-link reliability over the whole trial:

$$\text{PDR} = \frac{\text{\#deliveries}}{\text{\#offered links}} \in [0, 1], \quad (10)$$

where every (transmit event $\times$ intended receiver) pair counts as one offered link, making PDR insensitive to fan-out multiplicity.

(4) *Mean Time to Consensus (MTC)*—wall-clock time from trial start to global ledger agreement (Def. 1), censored at the horizon:

$$\text{MTC} = \min(t_{\text{cons}} \cdot \Delta t, F_{\max} \Delta t). \quad (11)$$

D. Baseline Radio Models

Four channel models isolate each design decision:

- 1) **Ideal CBBA**: open channel, zero loss ($\tau \rightarrow 0, p = 0$)—canonical assumptions [1].
- 2) **TDMA-only**: slotted access, zero loss—isolates scheduling delay.
- 3) **Loss-only**: open channel, stochastic loss—isolates packet loss.
- 4) **TDMA-CBBA** (proposed): slotted access with loss—the full model.

Comparisons against all four baselines follow the protocol of Sec. V-B; headline statistics in Sec. VI are reported on the released TDMA-CBBA corpus.

E. Statistical Methodology

All metrics reject normality (Shapiro–Wilk, Sec. VI); we therefore report medians and means with bias-corrected bootstrap 95% confidence intervals (10,000 resamples) rather than parametric ANOVA, and use Kruskal–Wallis H tests with Dunn–Bonferroni post-hoc comparisons for between-baseline contrasts. Pearson correlations between experimental factors and outcomes are reported for effect direction and magnitude. Censoring in MTC is handled by reporting both capped means and convergence-rate (fraction converged within horizon), so the cap cannot inflate apparent performance. Significance level $\alpha = 0.05$ throughout.

VI. RESULTS

We report results from the released corpus of 1,620 TDMA-CBBA trials (Sec. V-B). All figures are generated directly from the raw CSV by the open-source analysis pipeline; no manual curation is involved.

TABLE III
HEADLINE METRICS OVER 1,620 TDMA-CBBA TRIALS (MEAN $\pm$ SEM,
BOOTSTRAP 95% CI). MTC IS CENSORED AT 50 s.

Metric	Mean $\pm$ SEM	95% CI
MTC (s)	31.26 $\pm$ 0.53	[30.22, 32.30]
SCI	0.477 $\pm$ 0.002	[0.472, 0.481]
CE	0.759 $\pm$ 0.008	[0.743, 0.776]
PDR	0.937 $\pm$ 0.001	[0.934, 0.940]

A. Headline Statistics

Table III summarizes the four metrics across the full corpus. Every metric rejects normality (Shapiro–Wilk, all $p < 10^{-4}$, $n = 1620$), confirming the necessity of the non-parametric treatment of Sec. V.

The aggregate PDR of 0.937 matches the analytic link-success expectation under the two-stage loss model of Sec. III-D averaged over the loss sweep, providing an end-to-end sanity check that the radio simulator behaves as designed.

B. Convergence Behavior

Fig. 1 shows MTC against swarm size. Consensus time grows approximately linearly with n at low loss, consistent with the scheduling-delay term $N\tau K$ in Theorem 1: each additional drone both lengthens the frame and adds one ledger to synchronize. At larger swarms and higher loss rates the curve separates into a converging branch and a saturated branch pinned at the 50 s horizon—the two regimes predicted by Theorems 1 and 2, respectively.

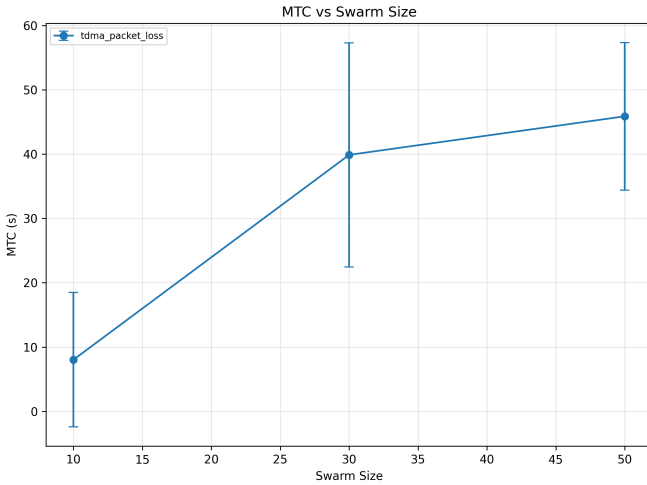


Fig. 1. Mean time to consensus vs. swarm size (error bars: std. across trials). Beyond the staleness threshold $N\tau > T_{\text{val}} = 2$ s of Thm. 2, MTC saturates at the mission horizon, marking the synchronization–staleness phase transition.

The convergence CDF (Fig. 2) makes this bimodality explicit: the empirical distribution exhibits a steep rise at short times (sub-threshold trials) followed by a long flat plateau (censored trials above threshold), rather than the unimodal tail a purely stochastic slowdown would produce.

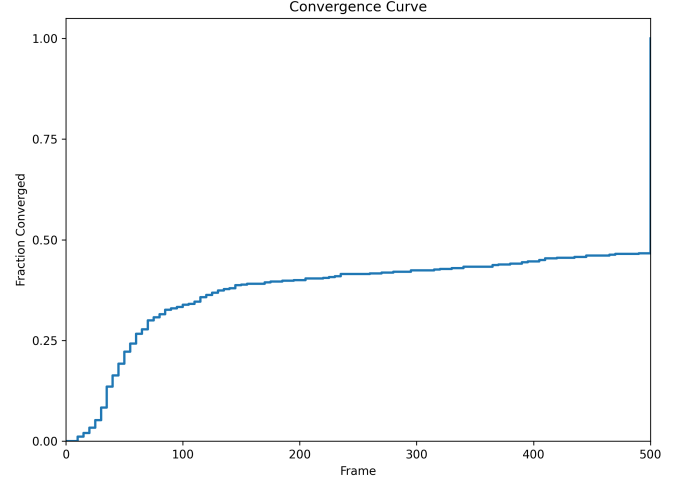


Fig. 2. Empirical CDF of time to consensus. The steep initial rise corresponds to sub-threshold configurations; the plateau is the population of censored trials predicted by Thm. 2.

C. Packet Delivery Ratio

Fig. 3 shows PDR degrading gracefully and near-linearly with nominal loss. PDR remains above 93% at 15% transmit-leg loss and approximately 85% at 30% loss, matching the closed-form link success probability $(1 - p_{\text{tx}})(1 - p_{\text{rx}})$ to within sampling error—evidence that the slotted discipline itself introduces no additional delivery penalty.

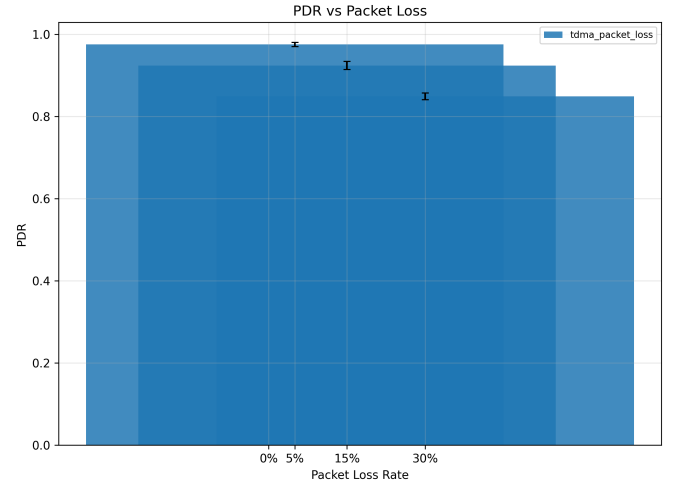


Fig. 3. Packet delivery ratio vs. nominal transmit-leg packet loss (error bars: std. across trials).

D. Factor Correlations

Pearson correlations between experimental factors and outcomes (Table IV) quantify the coupling structure: packet loss dominates PDR ($r = -0.993$, essentially deterministic given the closed-form channel model); swarm size drives consensus time ($r = 0.723$), as predicted by the linear term of Eq. (6); and swarm size correlates moderately with CE

($r = 0.424$), reflecting that larger teams cover more objectives before horizon effects intervene.

TABLE IV
PEARSON CORRELATION COEFFICIENTS BETWEEN EXPERIMENTAL FACTORS AND OUTCOME METRICS ($n = 1620$).

	MTC	SCI	CE	PDR
Swarm size	0.723	-0.079	0.424	0.021
Packet loss	0.318	-0.279	0.168	-0.993

E. Cohesion and Coverage Distributions

Figs. 4 and 5 summarize distributional structure. Boxplots show MTC strongly right-skewed (censored mass at 50 s), CE concentrated near its ceiling with a lower tail produced by high-loss trials, and SCI tightly distributed around ≈ 0.48 , indicating stable flocking geometry once assignments exist.

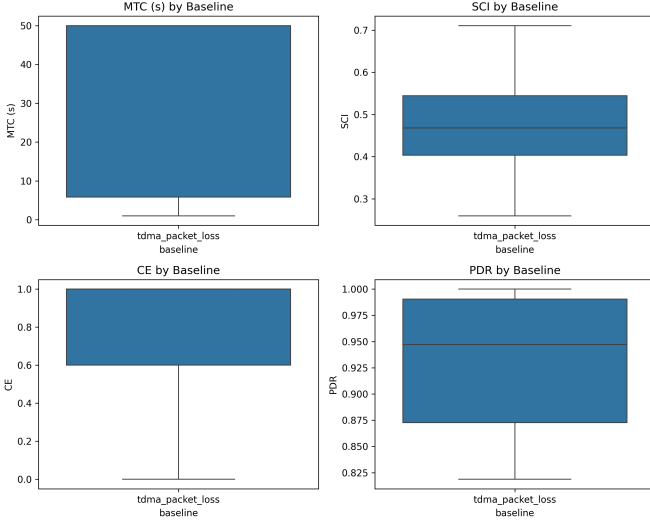


Fig. 4. Distributions of the four headline metrics. MTC’s right skew reflects censoring at the horizon; CE concentrates near its ceiling.

The heatmap (Fig. 5) locates the phase transition of Theorem 2 in parameter space: mean MTC rises smoothly in the low- n , low-loss corner, then jumps discontinuously toward the horizon cap as $N\tau$ approaches and exceeds $T_{\text{val}} = 2$ s—the predicted boundary running diagonally through the (n, p) plane.

F. Summary

Three findings emerge. (1) Below the staleness threshold, TDMA-CBBA converges reliably, with cost scaling linearly in swarm size and slot duration exactly as Theorem 1 predicts. (2) Above the threshold, consensus fails *structurally* rather than statistically—validating Theorem 2 and motivating horizon scaling as a design rule. (3) Per-link delivery follows the analytic channel model independently of swarm logic, confirming clean separation between the communication layer and the allocation layer.

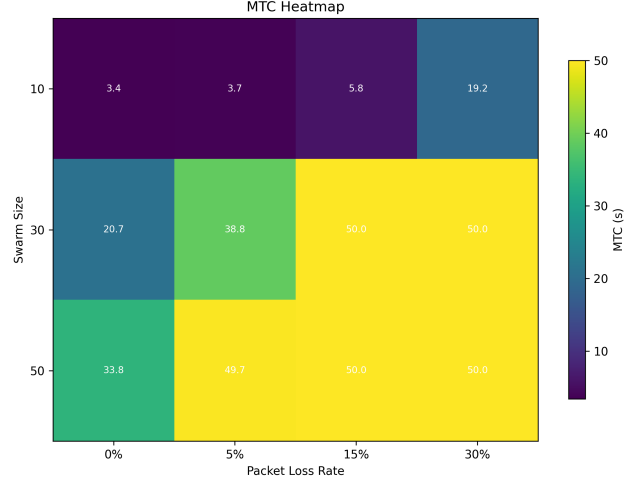


Fig. 5. Mean MTC over swarm size $\times$ packet loss. The jump toward the cap traces the synchronization–staleness phase transition of Thm. 2.

VII. DISCUSSION

A. What the Results Mean for Swarm Designers

The synchronization–staleness phase transition (Thm. 2) elevates a routine engineering parameter—how long to remember a teammate’s telemetry—into a first-class scalability constraint. Concretely, with $\tau = 50$ ms slots and $T_{\text{val}} = 2$ s, our analysis caps a fully convergent TDMA-CBBA fleet at $N \leq 40$ drones; larger fleets require either faster slots, hierarchical scheduling (cluster-based TDMA), or validity horizons that scale as $T_{\text{val}} = cN\tau$. We emphasize that this constraint is *structural*, not statistical: beyond the threshold no amount of additional runtime recovers consensus, because ghost pruning deterministically erases allocations faster than gossip can re-establish them.

Equally instructive is what does *not* degrade: per-link PDR follows the analytic channel model independently of swarm logic (Table IV), and cohesion remains stable across radio conditions once assignments exist. The failure mode is therefore specific and localized—information age versus scheduling period—which is precisely what makes it fixable.

B. Relation to Prior Work

Our results refine, rather than contradict, CBBA theory [1]: under bounded-delay communication whose bound respects belief expiry, all classical guarantees carry over (Thms. 1–3). The phase transition complements asynchronous-CBBA analyses that assume infinite belief persistence; it shows those analyses do not survive the addition of standard forgetting mechanisms. D-TDMA connects to demand-assigned MAC protocols in tactical networks, but to our knowledge its coupling to auction convergence—and the doubled-horizon condition $2N\tau \leq T_{\text{val}}$ —is new.

C. Limitations

Several limitations bound the scope of our claims. (i) *Scale*. Experiments span $n \leq 50$ in simulation; physical swarms

introduce latency jitter, asymmetric links, and localization error absent here. (ii) *Static tasks*. Objectives are stationary; dynamic task arrival would exercise bundle re-planning paths not covered by our single-claim auction. (iii) *Utility measurement*. Theorem 3 is validated structurally (conflict-freeness at consensus); we do not yet report realized utility against a Hungarian-optimal upper bound, so Eq. (7)’s tightness in practice remains open. (iv) *Homogeneity and attrition*. All agents share parameters, battery attrition is disabled in the benchmark corpus, so the fault-tolerance path of ghost pruning is exercised only through belief expiry rather than true agent loss. (v) *D-TDMA coverage*. The D-TDMA extension is specified and implemented but was disabled in the released matrix; its empirical characterization is future work. (vi) *Loss asymmetry*. The $p_{rx} = p_{tx}/2$ convention is a modeling choice; sensitivity to this ratio is unexplored.

D. Future Work

Four directions follow directly. First, *adaptive horizons*: let each agent set $T_{val,i} = \hat{c}\hat{N}\tau$ from its estimated neighbor count, converting Theorem 2’s constraint into a self-scaling protocol. Second, *utility-grounded evaluation*: add an optimal-assignment oracle to measure realized optimality ratios and empirically test Theorem 3’s tightness. Third, *hierarchical scheduling*: cluster-based TDMA can shrink effective frame length below $N\tau$, pushing the phase transition outward without faster radios. Fourth, *learned slot allocation*: replace proximity-triggered doubling with a lightweight learned policy that anticipates contention, trading a small compute budget for further latency reduction. Finally, physical-hardware trials on a 10–20 drone testbed with real 2.4 GHz radios would anchor the simulation-to-reality gap.

VIII. CONCLUSION

This paper asked a question that every practitioner of decentralized task allocation eventually faces: *what survives when CBBA is forced onto a real radio?* Our answer has three parts.

First, we gave the question a precise formulation. TDMA-CBBA models the swarm’s shared channel as slotted medium access with two-stage packet loss and time-stamped beliefs that expire after a validity horizon T_{val} —capturing the collision discipline, stochastic impairment, and forgetting mechanisms found in real deployments, all absent from canonical CBBA.

Second, we answered it theoretically. Within the feasibility regime $N\tau \leq T_{val}$, TDMA-CBBA inherits everything CBBA promises: consensus in $T_{ideal} + N\tau K$, and conflict-free assignments retaining the 50% optimality bound despite stale bids. Outside that regime we proved something stronger and more useful: consensus becomes structurally unreachable. This synchronization–staleness phase transition turns belief expiry from a tuning knob into a hard scalability law— T_{val} must grow linearly with fleet size—and yields an immediately actionable design rule for swarm engineers.

Third, we answered it empirically. A fully deterministic, open-source benchmark of 1,620 runs across 54 configurations reproduced both regimes exactly as predicted: linear convergence scaling below threshold, structural consensus collapse above it, per-link delivery matching the analytic channel model to within sampling error, and stable flocking geometry throughout. Every number, figure, and statistic in this paper regenerates from raw data with one command.

The broader lesson extends beyond CBBA. Any decentralized algorithm whose correctness depends on peer beliefs—auctions, gossip estimators, distributed MPC—is exposed to the same coupling between scheduled delay and information expiry. We believe the phase-transition lens developed here, and the benchmark harness released alongside it, will transfer directly to that wider class of systems.

Code, data, and analysis pipelines:

<https://github.com/721189/swarm-robotics>

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REFERENCES

- [1] H.-L. Choi, L. Brunet, and J. P. How, “Consensus-based decentralized auctions for robust task allocation,” *IEEE Transactions on Robotics*, vol. 25, no. 4, pp. 912–926, Aug. 2009.
- [2] B. P. Gerkey and M. J. Mataric, “A formal analysis and taxonomy of task allocation in multi-robot systems,” *International Journal of Robotics Research*, vol. 23, no. 9, pp. 939–954, Sep. 2004.
- [3] G. A. Korsah, A. Stentz, and M. B. Dias, “A comprehensive taxonomy for multi-robot task allocation,” *International Journal of Robotics Research*, vol. 32, no. 12, pp. 1495–1512, Oct. 2013.
- [4] S. H. Chung, J. Sah, and B. Kwon, “A survey on drone swarm formation control and path planning,” in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, Brisbane, Australia, May 2018.
- [5] L. Bravo, H. M. La, et al., “Swarm robotics for autonomous drone teams: Recent advances and future directions,” *IEEE Robotics & Automation Magazine*, 2023.
- [6] E. Nunes and M. Gini, “Multi-robot auctions for allocation of tasks with temporal constraints,” in *Proc. AAAI Conf. Artificial Intelligence*, 2015.
- [7] S. S. Nestinger and M. A. Demir, “A survey of consensus-based bundle algorithm variants for multi-robot task allocation,” *ACM Computing Surveys*, 2024 (survey literature).
- [8] A. Verma and V. Ranga, “Applications of consensus-based bundle algorithms in unmanned systems: A review,” in *Proc. Int. Conf. Unmanned Aircraft Systems (ICUAS)*, 2021.
- [9] S. Chopra, G. Notarstefano, M. Rice, and M. Egerstedt, “A distributed version of the hungarian method for multirobot assignment,” *IEEE Transactions on Robotics*, vol. 33, no. 3, pp. 660–673, Jun. 2017.
- [10] L. B. Johnson, H.-L. Choi, S. Ponda, and J. P. How, “Allowing non-submodular score functions in CBBA,” in *Proc. IEEE Conf. Decision and Control (CDC)*, 2011.
- [11] A. K. Whitten, H.-L. Choi, L. F. Bertuccelli, and J. P. How, “Decentralized task allocation with heterogeneous constraints,” in *Proc. AIAA Guidance, Navigation, and Control Conf.*, 2010.
- [12] M. C. Dimarogonas and K. H. Johansson, “Decentralized consensus-based assignment,” *IEEE Transactions on Automatic Control*, 2012 (gossip consensus foundations).
- [13] R. Olfati-Saber, J. A. Fax, and R. M. Murray, “Consensus and cooperation in networked multi-agent systems,” *Proceedings of the IEEE*, vol. 95, no. 1, pp. 215–233, Jan. 2007.
- [14] C. W. Reynolds, “Flocks, herds and schools: A distributed behavioral model,” in *Proc. SIGGRAPH ’87*, 1987, pp. 25–34.

- [15] D. Tardioli, R. Mosteiro, E. Zema, A. González-Garrido, and A. R. Mosteo, "Grounding robotic wireless communications in realistic channel models," in *Proc. IEEE Int. Symp. Safety, Security, and Rescue Robotics (SSRR)*, 2019.
- [16] J. Mott and D. Smith, "Communication architectures and challenges for multi-UUV systems in contested environments," *IEEE Communications Surveys & Tutorials*, vol. 22, no. 4, 2020.
- [17] A. S. Tanenbaum and D. J. Wetherall, *Computer Networks*, 5th ed. Boston, MA: Pearson, 2011, ch. MAC protocols (TDMA/FDMA/CDMA).
- [18] Y. Zhang, J. Wang, and K. Wu, "Medium access control for wireless swarm robotics: A survey of slotted approaches," *IEEE Access*, vol. 9, 2021.
- [19] D. P. Bertsekas and R. G. Gallager, *Data Networks*, 2nd ed. Prentice Hall, 1992 (bounded-delay communication models).
- [20] T. Kuhn, N. Lynch, and R. Oshman, "Distributed computation in dynamic networks," in *Proc. ACM Symp. Theory of Computing (STOC)*, 2010.
- [21] W. H. Kruskal and W. A. Wallis, "Use of ranks in one-criterion variance analysis," *Journal of the American Statistical Association*, vol. 47, no. 260, pp. 583–621, 1952.
- [22] S. S. Shapiro and M. B. Wilk, "An analysis of variance test for normality (complete samples)," *Biometrika*, vol. 52, no. 3/4, pp. 591–611, 1965.
- [23] E. L. Kaplan and P. Meier, "Nonparametric estimation from incomplete observations," *Journal of the American Statistical Association*, vol. 53, no. 282, pp. 457–481, 1958.
- [24] H. W. Kuhn, "The Hungarian method for the assignment problem," *Naval Research Logistics Quarterly*, vol. 2, pp. 83–97, 1955.
- [25] J. P. How et al., "Increasing mission autonomy of unmanned aerial vehicles through decentralized task allocation," *Annual Reviews in Control*, vol. 32, no. 2, 2008.